Department of Electrical and Information Engineering
Faculty of Engineering, UOR
Hapugala, Galle, Sri Lanka

EE8203 - Design and Management of Data Networks

Group Project

Department-Level Campus Network Design & Management

Module	EE8203 - Design and Management of Data Networks
Assignment type	Group Project (2 students per group)
Total marks	20 marks (contributes to module continuous assessment)
Duration	9 weeks (refer to schedule in Section 5)
Simulation tool	GNS3 (Cisco IOSv / IOS images as available in the lab)

Academic integrity notice: All submitted work must be your own. Code and configuration must be demonstrated live. Plagiarism, including the uncritical use of AI-generated code without understanding, will be penalised.

1. Introduction and Scenario

The Faculty of Engineering at the University of Ruhuna, Hapugala premises, operates four academic and administrative departments:

Code	Department	Role in this scenario
DEIE	Dept. of Electrical & Information Engineering	Engineering research and undergraduate labs
DCEE	Dept. of Civil & Environmental Engineering	Administration and project coordination offices
DMME	Dept. of Mechanical & Manufacturing Engineering	Workshop lab and server access zone
DIS	Dept. of Inter Disciplinary Studies	Faculty IT hub - hosts central server farm

You are engaged as a network engineering team to design, implement, automate, and monitor the campus data network that interconnects all four departments. The network must enforce strict access control policies between departments, be configurable through automation scripts, and be continuously monitored through a network management system.

All configuration and testing are to be performed inside GNS3 using Cisco IOS router and switch images. A Linux virtual machine within the GNS3 topology serves as the automation and monitoring host.

2. Learning Objectives

On successful completion of this project, students will be able to:

- Design a hierarchical three-layer campus network topology with VLANs, trunking, and inter-VLAN routing
- Formulate and apply Access Control Lists (ACLs) to enforce inter-department security policies
- Automate device configuration programmatically using Python and the Netmiko library
- Write and execute Ansible playbooks for idempotent, repeatable switch configuration
- Justify the selection of network devices and automation tools with engineering rationale
- Deploy and configure Zabbix as a network monitoring and alerting platform
- Produce a professional Method of Procedure (MOP) document following industry conventions
- Demonstrate working knowledge through live testing and code walkthrough sessions

3. Network Design Requirements

3.1 Topology Architecture

The network must follow a three-layer hierarchical design. The minimum required device inventory is shown below. Students may add devices with justification, removing devices requires written approval.

Device / hostname	Qty	GNS3 image	Primary role
R-CORE	1	Cisco 7200 / IOSv	Core router, OSPF area 0, NAT
R-EDGE	1	Cisco 7200 / IOSv	Edge / WAN router, default route, NAT overload
SW-CORE	1	IOSvL2 / EtherSwitchRouter	Core L3 switch, inter-VLAN routing backbone
SW-D-DEIE, SW-D-DCEE, SW-D-DMME	3	IOSvL2	Distribution L3 switches, one per dept (HSRP optional)
SW-A-DEIE, SW-A-DCEE, SW-A-DMME, SW-A-DIS	4	IOSvL2	Access L2 switches, one per department
VM-AUTO	1	Linux (Ubuntu 22.04)	Automation host Python/Netmiko + Ansible + SSH
VM-ZABBIX	1	Linux (Ubuntu 22.04)	Zabbix server, SNMPv2c monitoring
VM-DHCP / VM-WEB	1	Linux	DHCP/DNS server; optional web server for reachability tests
VPC end-hosts	8+	GNS3 VPCs	Two per department — used for ACL policy testing

Total configurable network devices: Approximately 13 (2 routers + 8 switches + 3 VMs). Students must include a device selection justification section in their MOP and final report.

3.2 VLAN and IP Addressing Scheme

Students must design a complete VLAN and IP addressing plan. The minimum VLAN allocation is as follows. The IP subnets below are indicative, students may use any private ip address space but must document the full plan.

VLAN ID	Name	Department	Suggested subnet	Purpose
10	VLAN_DEIE	DEIE	10.10.10.0 /24	Engineering workstations
20	VLAN_DCEE	DCEE	10.10.20.0 /24	Civil/Environmental dept hosts
30	VLAN_DMME	DMME	10.10.30.0 /24	Mechanical/Manufacturing hosts
40	VLAN_DIS	DIS — Server Farm	10.10.40.0 /24	Faculty IT hub / server zone
99	MGMT	All devices	10.99.99.0 /24	Out-of-band management (SSH only)
100	NATIVE	Trunk links	N/A	Native VLAN (untagged frames)

3.3 Routing Protocol

- OSPF (single area, area 0) must be used for dynamic routing between routers and Layer-3 switches.
- Static default route on R-EDGE pointing to the simulated internet cloud.
- NAT overload (PAT) on R-EDGE - only VLAN_DEIE and VLAN_DCEE are permitted internet egress.

3.4 Access Control Policy Requirements

The following inter-department ACL policies must be implemented and verified. All policies are to be applied as extended named ACLs. Students must document the placement (interface, direction) of each ACL.

Source	Destination	Action	Policy rationale
VLAN_DEIE	VLAN_DIS	PERMIT all	Engineering staff have full server farm access
VLAN_DCEE	VLAN_DIS	PERMIT HTTP/HTTPS only	Admin staff may access web services in DIS only
VLAN_DMME	VLAN_DIS	DENY all	Workshop lab has no server farm access
VLAN_DCEE	VLAN_DEIE	DENY all	Admin must not access engineering subnet
VLAN_DMME	VLAN_DCEE	DENY all	Workshop is fully isolated from admin
VM_ZABBIX IP	All devices	PERMIT SNMP (UDP 161)	Monitoring agent must reach all devices

Source	Destination	Action	Policy rationale
MGMT VLAN only	All devices	PERMIT SSH (TCP 22)	SSH management restricted to VLAN 99
Any	Any	DENY all (implicit)	Default deny - must be explicitly acknowledged

Important: Students must produce a 4x4 department reachability test matrix (DEIE, DCEE, DMME, DIS) showing the result of ping and traceroute tests between every pair. Screenshot evidence is required for every cell.

4. Network Automation Requirements

4.1 Netmiko - Python Automation (Routers)

Students must write Python scripts using the Netmiko library to automate the configuration of R-CORE and R-EDGE, as well as the SNMP configuration push to all devices. Scripts must satisfy all of the following requirements:

- Use a device inventory file in JSON or YAML format - device parameters must not be hardcoded in scripts
- Connect to each device via SSH on VLAN 99 (management plane)
- Automate: interface IP addressing, OSPF configuration, NAT rules, and ACL deployment on routers
- Automate: SNMPv2c community string and trap destination configuration on all devices
- Include structured error handling (try/except) and write output to a timestamped log file
- Scripts must be commented and readable - marks are awarded for code quality, not just functionality
- Demonstrate idempotency where possible (re-running the script must not create duplicate config)

4.2 Ansible - Playbook Automation (Switches)

Students must write Ansible playbooks using the cisco.ios collection to automate the configuration of all distribution (SW-D-*) and access (SW-A-*) switches. The following requirements apply:

- Organise the project using Ansible roles - separate roles for VLANs, trunking, access ports, and STP
- Use **group_vars** and **host_vars** directories for device-specific and group-level variables
- Playbooks must be idempotent - running ansible-playbook --check must report no changes after initial application
- All playbooks must be version-controlled in a Git repository with meaningful commit messages
- Include a site.yml top-level playbook that orchestrates all roles in the correct order
- Demonstrate a rollback playbook that can restore a device to a clean baseline within 5 minutes

4.3 Tool Selection Justification

Students must include a written section (in both the MOP and the final report) justifying why Netmiko was selected for routers and Ansible for switches. The justification must address: the nature of the tasks, the scalability of each approach, idempotency considerations, and what limitations each tool has in this scenario.

5. Network Monitoring with Zabbix

A Zabbix 6.x LTS server must be deployed on VM-ZABBIX within the GNS3 topology. The monitoring system must meet the following minimum requirements:

5.1 Host Onboarding

- All routers and switches must be added as monitored hosts in Zabbix
- Use SNMPv2c as the monitoring interface (community string configured via automation in Section 4.1)
- Apply the appropriate built-in template: Template Net Cisco IOS SNMPv2 for all Cisco devices

5.2 Triggers and Alerts

The following triggers must be configured and demonstrated:

- Device unreachable (ICMP ping timeout > 3 consecutive polls)
- Interface down - alert within one polling cycle when any monitored interface transitions to down state
- High CPU utilisation - trigger when CPU > 80% for more than 60 seconds
- At least one custom trigger defined by the student (must be justified in the report)

5.3 Dashboard

- Create a custom Zabbix dashboard named **'FoE-UoR Network'** showing at minimum: host availability map, interface traffic graphs for core devices, and open trigger count
- Export and submit the dashboard configuration as a Zabbix JSON export file

6. Project Schedule and Deliverables

Week(s)	Phase	Activities	Deliverable
1	Design	IP addressing plan, VLAN scheme, topology diagram, device selection justification. GNS3 project skeleton.	Design document + topology diagram (PDF)
2–3	Baseline config	Manual configuration of all devices: VLANs, trunks, inter-VLAN routing, OSPF, management VLAN 99, SSH. Verify full reachability before automation.	GNS3 saved snapshot + connectivity test log
4	ACL implementation	Design and apply all inter-department ACLs. Test using ping/traceroute from VPCs. Populate 4×4 reachability matrix.	ACL policy table + test evidence screenshots
5–6	Netmiko (Python)	Write Python scripts for router + SNMP automation. Code review session in week 6 lab, be prepared to explain every function.	Python scripts + execution log. Code review in lab.
7–8	Ansible (switches)	Write Ansible playbooks with roles. Test idempotency. Commit all work to Git with meaningful history.	Ansible Git repo link. Playbook run output.
9	Zabbix monitoring	Deploy Zabbix. Onboard all devices via SNMP. Configure triggers and dashboard. Demonstrate trigger firing.	Dashboard screenshot + trigger fire evidence
9	MOP + final submission	Complete MOP document. Full reset-and-replay test. Record 10-minute demo video. Submit all artefacts.	All submission artefacts (see Section 8)

Checkpoint session (Week 5): The lecturer will conduct a brief mid-project checkpoint in week 5 to assess progress. Students who are significantly behind at this stage will be advised and may receive guidance sessions. Attendance is mandatory.

7. Assessment Criteria

The assignment is marked out of 20. The following rubric applies. All marks are subject to live demonstration - a submission that cannot be demonstrated will receive zero for the relevant component.

Component	Assessment criteria	Marks	Verification method
Network design & device justification	Complete and correct IP plan, VLAN scheme, topology diagram. Written justification for all device selections referencing OSI layer roles, scalability, and cost.	3	Design document + diagram
ACL implementation & policy correctness	All specified policies implemented correctly. Proper placement (inbound/outbound, correct interface). Complete 4×4 test matrix with screenshot evidence.	4	Live ACL test + matrix
Netmiko Python automation	Functional scripts that configure routers from a clean state. Code quality: error handling, logging, parameterised inventory, comments. Idempotency demonstrated. Student understands code.	4	Code review + reset demo
Ansible playbooks	Working playbooks with role structure. Idempotency verified (--check). Meaningful Git history. site.yml orchestration. Rollback playbook present.	4	Git log + playbook run
Zabbix monitoring	All devices onboarded. Required triggers configured and demonstrated firing. Custom dashboard present. Dashboard JSON exported.	3	Live Zabbix demo
MOP document quality	Completeness per Section 9 structure. Clarity and professional tone. Risk table present. Rollback procedures for each major change. Post-implementation review completed.	2	Document review
TOTAL		**20**	

7.1 Code Assessment Procedure

The following checks will be applied during the live demonstration session:

Check	What the assessor looks for	Method
Oral code walkthrough	Student explains each function/task in their Netmiko script and Ansible playbook. Cannot be passed by memorisation — questions are contextual.	Live, ~10 min
Reset and replay	One device is wiped. Automation scripts are run. Device must be fully reconfigured and reachable within 5 minutes.	Live demo
Git commit history	Commits span the full 10 weeks with meaningful messages. Single large commit on final day is a red flag for dishonesty.	Git log review
Idempotency test	Netmiko script run twice - no duplicate config. Ansible --check run after application - zero changes reported.	CLI output
Zabbix trigger fire	An interface is shut on a GNS3 device. Assessor verifies an alert appears in Zabbix within the polling interval.	Live demo
ACL policy test	Assessor selects 3 cells from the 4×4 matrix and verifies live using ping/traceroute.	Live from VPC

8. Submission Requirements

All artefacts must be submitted via the ELMS by the deadline communicated in the module schedule. Late submissions are penalised. The submission must include the following:

- Python scripts (Netmiko)
- Ansible project - a link to the Git repository (GitHub/GitLab)
- Zabbix dashboard JSON export (exported from Zabbix Configuration > Export)
- MOP document (PDF and editable .docx)
- Final technical report (PDF, maximum 20 pages excluding appendices) following the report structure in Appendix A
- 10-minute demonstration video (MP4, narrated, showing all required checks from Section 7.1)
- 4×4 reachability test matrix with screenshot evidence (included in the report appendix)

Naming convention: All submitted files must be named: EE8203_[StudentID1_ StudentID2]_[Component] - e.g. EE8203_3066_3067_MOP.pdf. Submissions not following this format may be returned unmarked.

9. Method of Procedure (MOP) - Required Structure

A Method of Procedure is a standard industry document used by network operations engineers to plan, execute, and recover from any network change. Producing one is a professional skill and is a graded component of this assignment. Your MOP must follow the structure below and be updated iteratively throughout the project.

MOP Section 1 - Document Header

- Project title, student name(s) and IDs, module code, version number, date, and lecturer name
- Estimated change window duration and target completion date

MOP Section 2 - Scope and Objectives

- What the MOP implements (full network build, phased)
- Devices in scope - list by hostname
- Explicitly state what is out of scope

MOP Section 3 - Prerequisites and Pre-checks

- GNS3 running and project loaded: verify with [specific check]
- All devices reachable on VLAN 99: verify with ping sweep from VM-AUTO
- Ansible inventory ping: ansible all -m ping - must return SUCCESS for all hosts
- Zabbix server accessible on port 80: verify with curl http://[Zabbix IP]/zabbix
- Each prerequisite must have a pass/fail check command documented

MOP Section 4 - Risk Assessment

Include a risk table with the following columns:

Risk	Likelihood	Impact	Mitigation
ACL blocks management SSH access	Medium	High	Apply ACL only to data VLANs; verify SSH on VLAN 99 before applying
Automation script overwrites existing config	Low	Medium	Take GNS3 snapshot before each automation run; test on one device first
[Student-defined risk]	...	...	...

MOP Section 5 - Step-by-Step Procedure

Each phase must have numbered steps. Every step must state: (a) the action, (b) the expected output, and (c) the verification command. Example format:

1. Phase 1 - Baseline Configuration
   - 1.1. Connect to SW-D-DEIE via SSH: ssh admin@10.99.99.11
   - 1.2. Create VLANs: vlan 10 / name VLAN_DEIE - Expected: VLAN appears in show vlan brief
   - 1.3. Verify: show vlan brief - confirm VLAN 10 active and assigned to correct ports

MOP Section 6 - Test Plan

- 4×4 department reachability matrix (populate after testing)
- Netmiko script execution log (attach as appendix)
- Ansible playbook run output (attach as appendix)
- Zabbix dashboard screenshot
- Zabbix trigger fire evidence (screenshot with timestamp)

MOP Section 7 - Rollback Procedure

- For each major change, document how to revert. Example: if ACL blocks management access — connect to R-CORE on console, issue: no ip access-group ACL_DEIE_IN in on the affected interface.
- Rollback for Netmiko errors: restore from GNS3 snapshot taken before script run
- Rollback for Ansible: use the rollback.yml playbook in the /playbooks/rollback/ directory

MOP Section 8 - Post-Implementation Review

- What worked as planned and what did not
- Deviations from the MOP - what changed and why
- Lessons learned - what you would do differently
- This section is written after the project is complete and contributes to the MOP quality mark

Appendix A — Final Report Structure

The final report must follow this structure. Maximum 20 pages excluding appendices. Use 12pt Arial, 1-inch margins.

2. Executive Summary (1 page)
3. Network Design (topology diagram, VLAN table, IP addressing plan, device justification)
4. ACL Policy Implementation (ACL design table, 4×4 reachability matrix with evidence)
5. Network Automation (Netmiko - approach, script description, execution evidence; Ansible role structure, idempotency evidence; tool selection justification)
6. Monitoring System (Zabbix setup, host list, triggers, dashboard description)
7. Challenges and Lessons Learned
8. References
9. Appendices (script listings, playbook listings, full test logs, Zabbix export)